EKS Session

Summary 30-04-2023

- Manage the route rules in an EKS (Elastic Kubernetes Service) cluster using Kubernetes (k8s) rather than AWS Cloud by deploying **Custom Resource Definitions (CRDs)**.

- CRDs allow you to define your own custom resources in Kubernetes. **Ingress CRDs allow you to define custom routing rules.**

- In Kubernetes, you can use the Ingress resource to manage route rules. An Ingress is a Kubernetes resource that provides a way to manage external access to services in a cluster. You can define rules in an Ingress that determine how traffic should be routed to different services based on the requested path or hostname.

```
C:\Users\Vimal Daga>kubectl  get  crd
NAME                                          CREATED AT
eniconfigs.crd.k8s.amazonaws.com              2023-04-29T05:45:28Z
ingressclassparams.elbv2.k8s.aws              2023-04-29T06:53:30Z
securitygrouppolicies.vpcresources.k8s.aws    2023-04-29T05:45:31Z
targetgroupbindings.elbv2.k8s.aws             2023-04-29T06:53:30Z
```

- **Ingress rules Example**: Launch 2 services with deployments.

  - Launch deployment for mymail.

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: mail
spec:
  replicas: 3
  selector:
    matchLabels:
      app: mymail
    matchExpressions:
      - { key: app, operator: In, values: [mymail] }
  template:
    metadata:
      name: mymail-pod
      labels:
        app: mymail
    spec:
      containers:
      - name: myweb-con
        image: vimal13/apache-webserver-php
```

```
C:\Users\Vimal Daga\Desktop\kube_code>notepad deploy.yml

C:\Users\Vimal Daga\Desktop\kube_code>kubectl apply -f deploy.yml
deployment.apps/mail created

C:\Users\Vimal Daga\Desktop\kube_code>kubectl ge tdeploy
error: unknown command "ge" for "kubectl"

Did you mean this?
        set
        get
        cp

C:\Users\Vimal Daga\Desktop\kube_code>kubectl get deploy
NAME           READY   UP-TO-DATE   AVAILABLE   AGE
mail           3/3     3            3           8s
myweb-deploy   3/3     3            3           22h
```

- Launch service for mymail deployment.

```yaml
apiVersion: v1
kind: Service
metadata:
  name: mymail-service
  labels:
    app: mymail-service
spec:
  type: NodePort
  selector:
    app: mymail
  ports:
    - port: 80
      targetPort: 80
      nodePort: 31001
```

```
C:\Users\Vimal Daga\Desktop\kube_code>notepad service.yml

C:\Users\Vimal Daga\Desktop\kube_code>kubectl apply -f service.yml
service/mymail-service created

C:\Users\Vimal Daga\Desktop\kube_code>kubectl get svc
NAME             TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
kubernetes       ClusterIP   10.100.0.1      <none>        443/TCP        24h
my-service       NodePort    10.100.76.202   <none>        80:31000/TCP   22h
mymail-service   NodePort    10.100.3.14     <none>        80:31001/TCP   11s
```

```
C:\Users\Vimal Daga\Desktop\kube_code>kubectl describe svc mymail-service
Name:                     mymail-service
Namespace:                default
Labels:                   app=mymail-service
Annotations:              <none>
Selector:                 app=mymail
Type:                     NodePort
IP Family Policy:         SingleStack
IP Families:              IPv4
IP:                       10.100.3.14
IPs:                      10.100.3.14
Port:                     <unset>  80/TCP
TargetPort:               80/TCP
NodePort:                 <unset>  31001/TCP
Endpoints:                192.168.5.45:80,192.168.61.170:80,192.168.88.218:80
Session Affinity:         None
External Traffic Policy:  Cluster
Events:                   <none>
```

- Launch deployment for mysearch.

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: search
spec:
  replicas: 3
  selector:
    matchLabels:
      app: mysearch
    matchExpressions:
      - { key: app, operator: In, values: [mysearch] }
  template:
    metadata:
      name: mysearch-pod
      labels:
        app: mysearch
    spec:
      containers:
      - name: myweb-con
        image: httpd
```

- Launch service for mysearch deployment.

```
apiVersion: v1
kind: Service
metadata:
  name: mysearch-service
  labels:
    app: mysearch-service
spec:
  type: NodePort
  selector:
    app: mysearch
  ports:
    - port: 80
      targetPort: 80
      nodePort: 31002
```

```
C:\Users\Vimal Daga\Desktop\kube_code>notepad deploy.yml

C:\Users\Vimal Daga\Desktop\kube_code>kubectl apply -f deploy.yml
deployment.apps/search created

C:\Users\Vimal Daga\Desktop\kube_code>notepad service.yml

C:\Users\Vimal Daga\Desktop\kube_code>kubectl apply -f service.yml
service/mysearch-service created
```

```
C:\Users\Vimal Daga\Desktop\kube_code>kubectl describe svc mysearch-service
Name:                     mysearch-service
Namespace:                default
Labels:                   app=mysearch-service
Annotations:              <none>
Selector:                 app=mysearch
Type:                     NodePort
IP Family Policy:         SingleStack
IP Families:              IPv4
IP:                       10.100.38.110
IPs:                      10.100.38.110
Port:                     <unset>  80/TCP
TargetPort:               80/TCP
NodePort:                 <unset>  31002/TCP
Endpoints:                192.168.40.110:80,192.168.5.57:80,192.168.80.78:80
Session Affinity:         None
External Traffic Policy:  Cluster
Events:                   <none>
```

- You can use annotations to specify the type of Application Load Balancer (ALB) that you want to use.

```
kind: Ingress
metadata:
  name: my-app-ingress
  annotations:
    alb.ingress.kubernetes.io/scheme: internet-facing
    alb.ingress.kubernetes.io/healthcheck-protocol: HTTP
    alb.ingress.kubernetes.io/healthcheck-port: traffic-port
    alb.ingress.kubernetes.io/healthcheck-path: /
    alb.ingress.kubernetes.io/healthcheck-interval-seconds: '15'
    alb.ingress.kubernetes.io/healthcheck-timeout-seconds: '5'
    alb.ingress.kubernetes.io/success-codes: '200'
    alb.ingress.kubernetes.io/healthy-threshold-count: '2'
    alb.ingress.kubernetes.io/unhealthy-threshold-count: '2'
spec:
```

- Rules of /mail path: That will go to mymail-service

```
spec:
  ingressClassName: awesome-class
  rules:
  - http:
      paths:
      - path: /mail
        pathType: Prefix
        backend:
          service:
            name: mymail-service
            port:
              number: 80
```

- Rules for /search path: That will go to mysearch-service

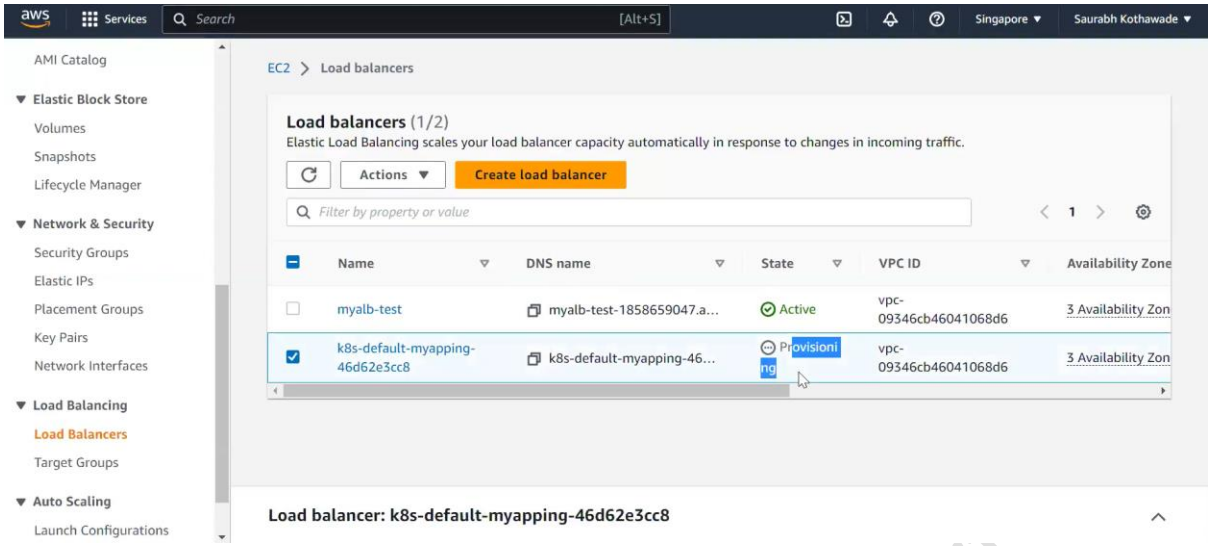
```
  - path: /search
    pathType: Prefix
    backend:
      service:
        name: mysearch-service
        port:
          number: 80
```

- Create an ingress rule.

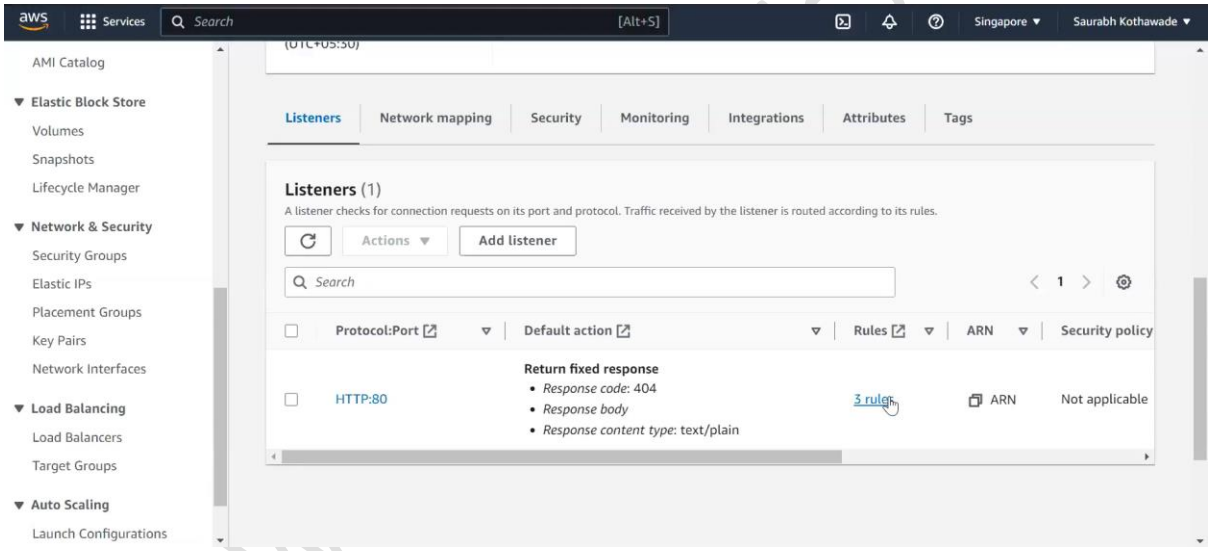
```
C:\Users\Vimal Daga\Desktop\kube_code>kubectl apply -f ingress-alb-rule.yml
ingress.networking.k8s.io/my-app-ingress created
```

```
C:\Users\Vimal Daga\Desktop\kube_code>kubectl get ingress
NAME              CLASS          HOSTS   ADDRESS                                                                 PORTS
   AGE
minimal-ingress   awesome-class   *       myalb-test-1858659047.ap-southeast-1.elb.amazonaws.com                  80
   22h
my-app-ingress    awesome-class   *       k8s-default-myapping-46d62e3cc8-754101094.ap-southeast-1.elb.amazonaws.com   80
   6s
```

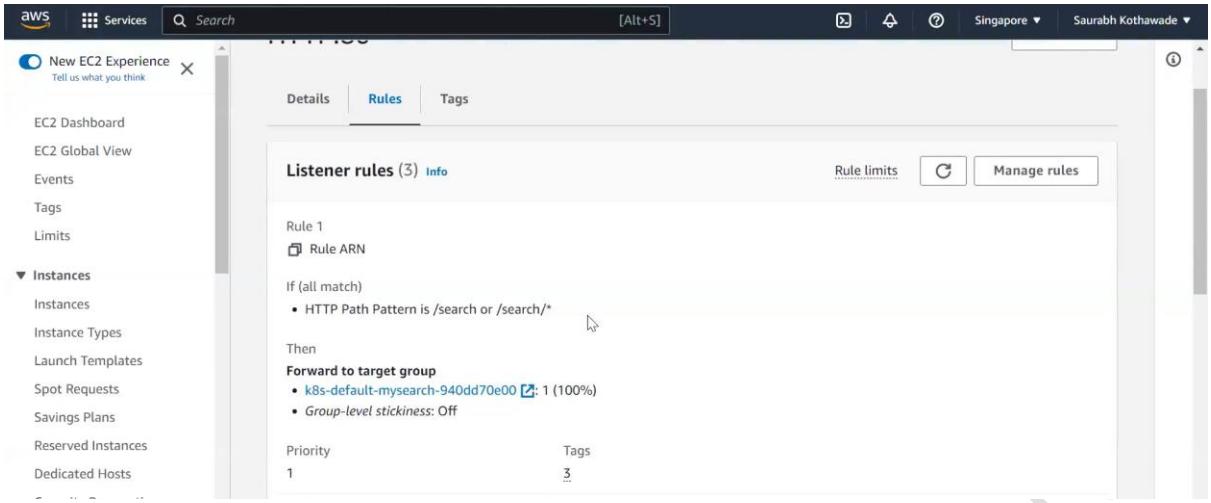
- After that, we can see that a load balancer will be immediately generated if we check the AWS console.



- In the AWS console, we can view the rules.



- Rule 1: For "/search"



- Rule 2: For "/mail"

Rule 2

Rule ARN

If (all match)

- HTTP Path Pattern is /mail or /mail/*

Then

Forward to target group

- k8s-default-mymailse-bceffff601: 1 (100%)
- *Group-level stickiness*: Off

Priority	Tags
2	3

- We will receive a 404 response code kind of page if we enter the incorrect path.

Default (last)

Rule ARN

If (all match)

- Request is not otherwise routed

Then

Return fixed response

- *Response code*: 404
- *Response body*
- *Response content type*: text/plain

Priority	Tags
default	0

- When you define an Ingress rule in Kubernetes, the path that you specify must match a valid URL path that can be handled by your application. For example, the "search" folder should be present in the web server's home directory if the path is "/search".

```
C:\Users\Vimal Daga\Desktop\kube_code>kubectl exec -it  mail-6f5676c969-tmcwq   -- bash
[root@mail-6f5676c969-tmcwq /]# cd /var/www/html/
[root@mail-6f5676c969-tmcwq html]# mkdir mail
[root@mail-6f5676c969-tmcwq html]# cd mail/
[root@mail-6f5676c969-tmcwq mail]# cat > index.html
i m mail 3
[root@mail-6f5676c969-tmcwq mail]# exit
exit
```

```
C:\Users\Vimal Daga\Desktop\kube_code>curl http://k8s-default-myapping-46d62e3cc8-754101094.ap-southeast-1.elb.amazonaws.com
/mail/index.html
i m mail 3

C:\Users\Vimal Daga\Desktop\kube_code>curl http://k8s-default-myapping-46d62e3cc8-754101094.ap-southeast-1.elb.amazonaws.com
/mail/index.html
i m mail 2

C:\Users\Vimal Daga\Desktop\kube_code>curl http://k8s-default-myapping-46d62e3cc8-754101094.ap-southeast-1.elb.amazonaws.com
/mail/index.html
i m mail 3
```

- Same applies for each single pod/container.

```
C:\Users\Vimal Daga\Desktop\kube_code>kubectl exec -it  search-79c45fbdc4-6n7tl   -- bash
root@search-79c45fbdc4-6n7tl:/usr/local/apache2# pwd
/usr/local/apache2
root@search-79c45fbdc4-6n7tl:/usr/local/apache2# ls
bin  build  cgi-bin  conf  error  htdocs  icons  include  logs  modules
root@search-79c45fbdc4-6n7tl:/usr/local/apache2# cd htdocs/
root@search-79c45fbdc4-6n7tl:/usr/local/apache2/htdocs# ls
index.html
root@search-79c45fbdc4-6n7tl:/usr/local/apache2/htdocs# mkdir search
root@search-79c45fbdc4-6n7tl:/usr/local/apache2/htdocs# cd search/
root@search-79c45fbdc4-6n7tl:/usr/local/apache2/htdocs/search# cat > index.html
i m search
root@search-79c45fbdc4-6n7tl:/usr/local/apache2/htdocs/search# exit
exit
```

- **Path types**:

1) Exact: Matches the URL path exactly and with case sensitivity.

2) Prefix: Matches based on a URL path prefix split by /

Examples

Kind	Path(s)	Request path(s)	Matches?
Prefix	/	(all paths)	Yes
Exact	/foo	/foo	Yes
Exact	/foo	/bar	No
Exact	/foo	/foo/	No
Exact	/foo/	/foo	No
Prefix	/foo	/foo , /foo/	Yes
Prefix	/foo/	/foo , /foo/	Yes
Prefix	/aaa/bb	/aaa/bbb	No
Prefix	/aaa/bbb	/aaa/bbb	Yes
Prefix	/aaa/bbb/	/aaa/bbb	Yes, ignores trailing slash
Prefix	/aaa/bbb	/aaa/bbb/	Yes, matches trailing slash

- **Host-based Ingress** rules in Kubernetes are used to route incoming requests based on the host name specified in the request's HTTP header.

```yaml
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: ingress-wildcard-host
spec:
  rules:
  - host: "foo.bar.com"
    http:
      paths:
      - pathType: Prefix
        path: "/bar"
        backend:
          service:
            name: service1
            port:
              number: 80
  - host: "*.foo.com"
    http:
      paths:
      - pathType: Prefix
        path: "/foo"
        backend:
          service:
            name: service2
            port:
              number: 80
```

- In Kubernetes, Ingress rules are processed in order, and the first rule that matches the incoming request is used to route the traffic. This means that

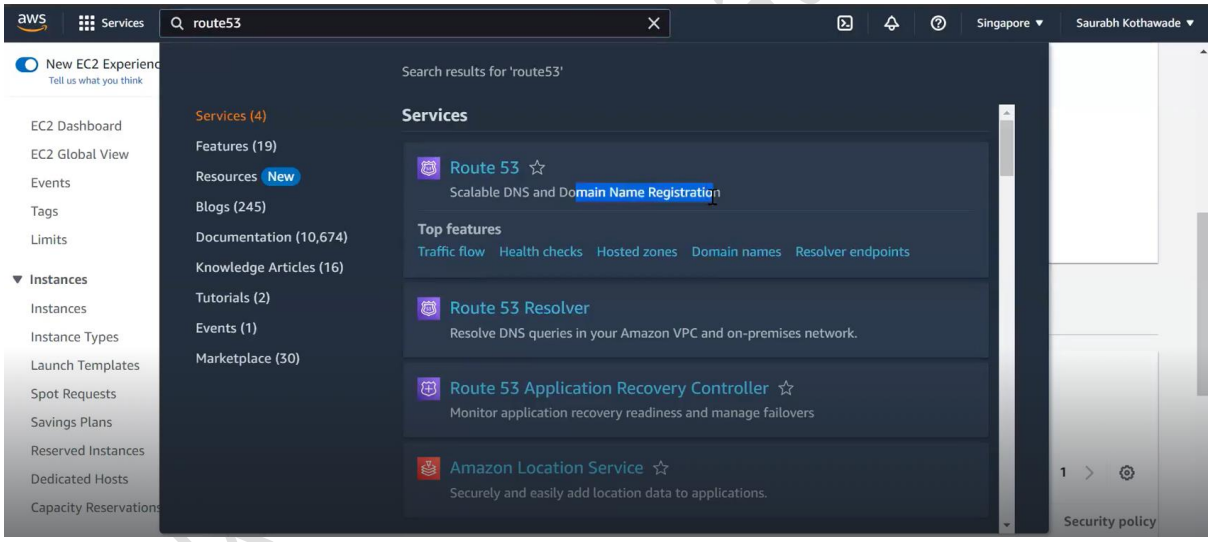
if you define a catch-all rule with a path of / at the beginning of your Ingress rule list, it will match all incoming requests, even if there are more specific rules defined later in the list.

- To avoid this issue, you should avoid defining a catch-all rule with a path of / at the beginning of your rule list. Instead, you can define a catch-all rule at the end of your list of rules.

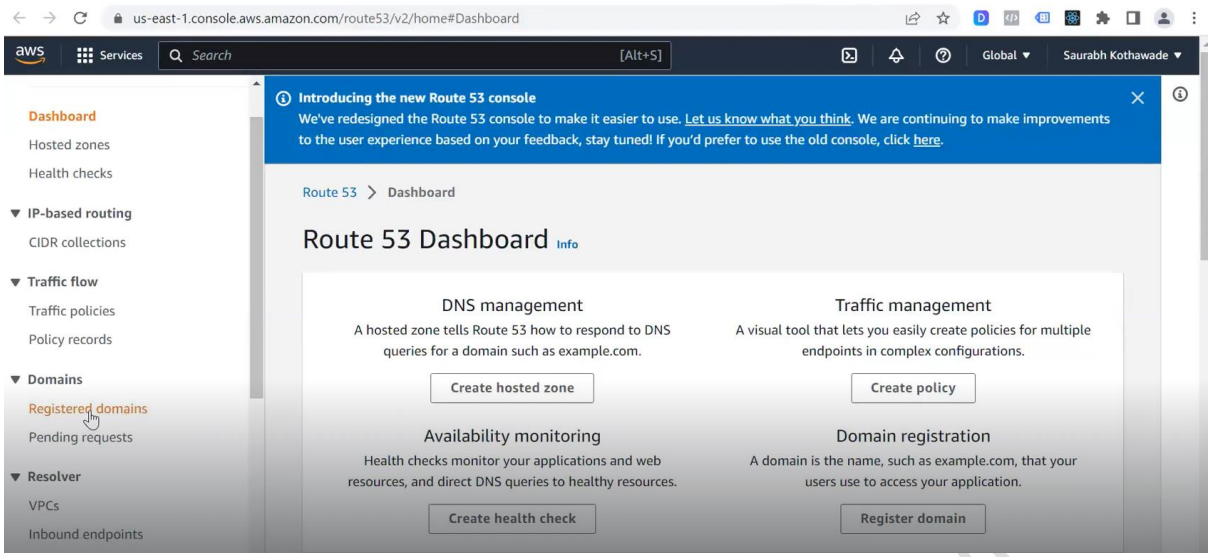
- **To convert an HTTP endpoint to HTTPS in an Amazon Elastic Kubernetes Service (EKS) cluster:**

- Before setting up HTTPS, you need to have a domain name for your service and an SSL certificate that matches the domain name:

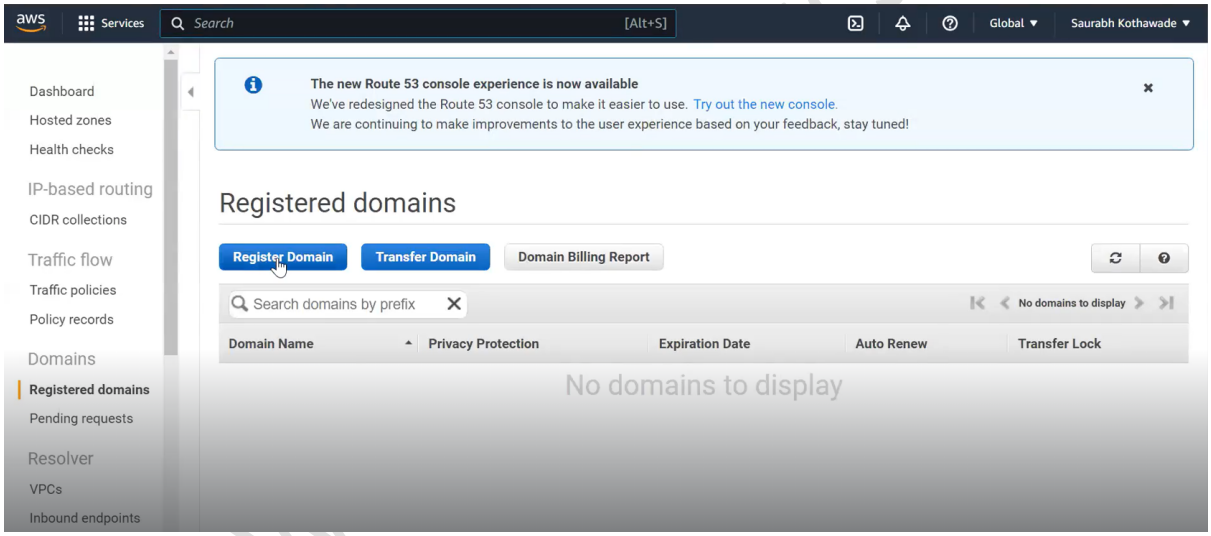
- Register a domain name:



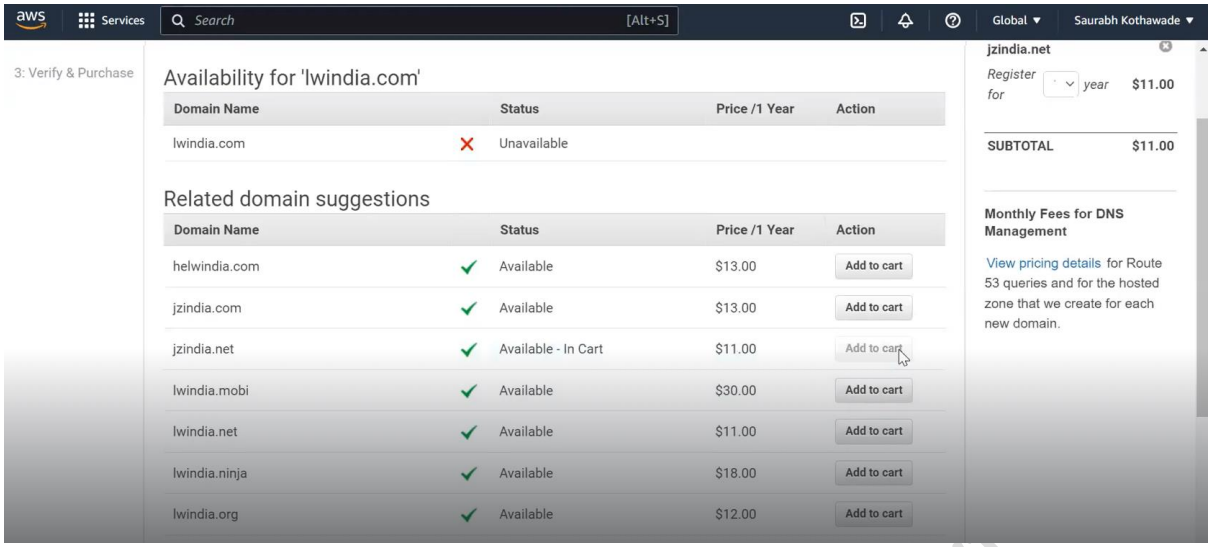
➢ Click on the "Registered Domains" option in the left-hand menu.



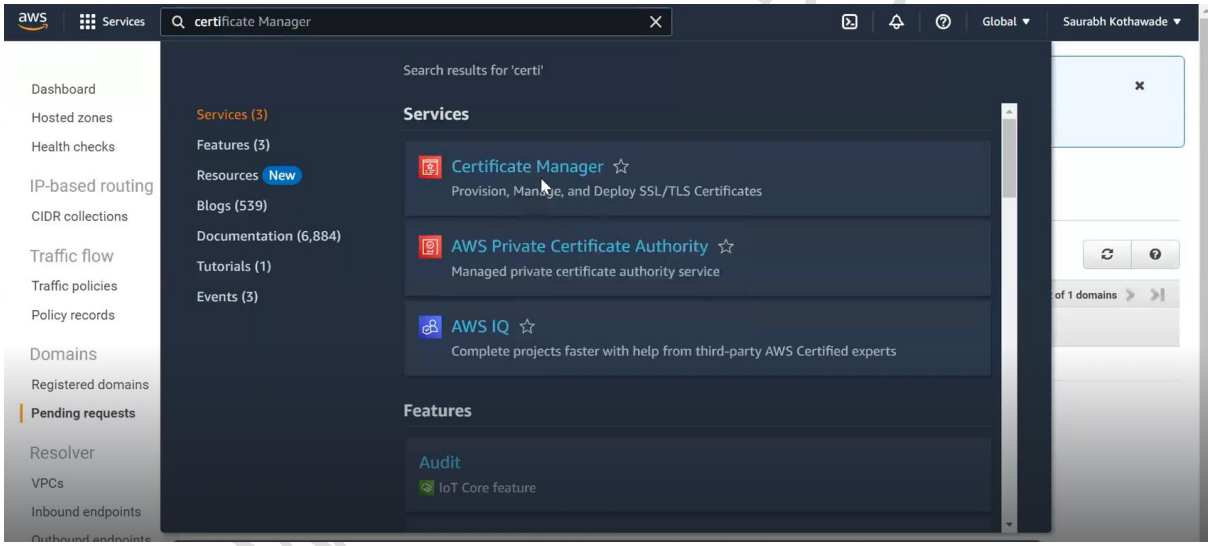
- Click on the "Register Domain" button.



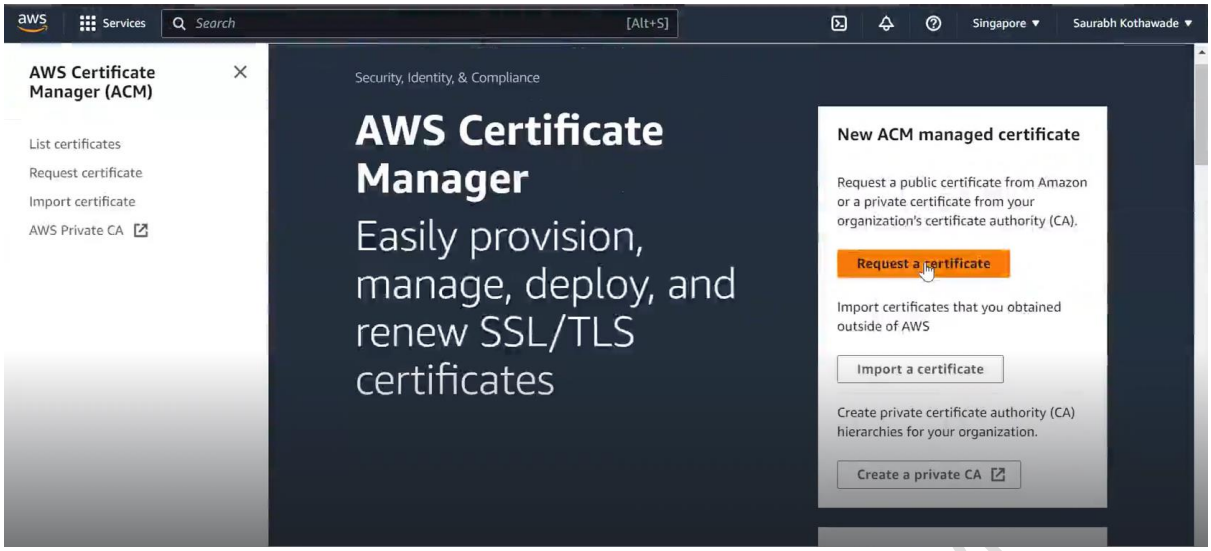
- Enter the domain name you want to register in the "Domain name" field.
- If the domain name is available, select it and click on the "Add to cart" button.

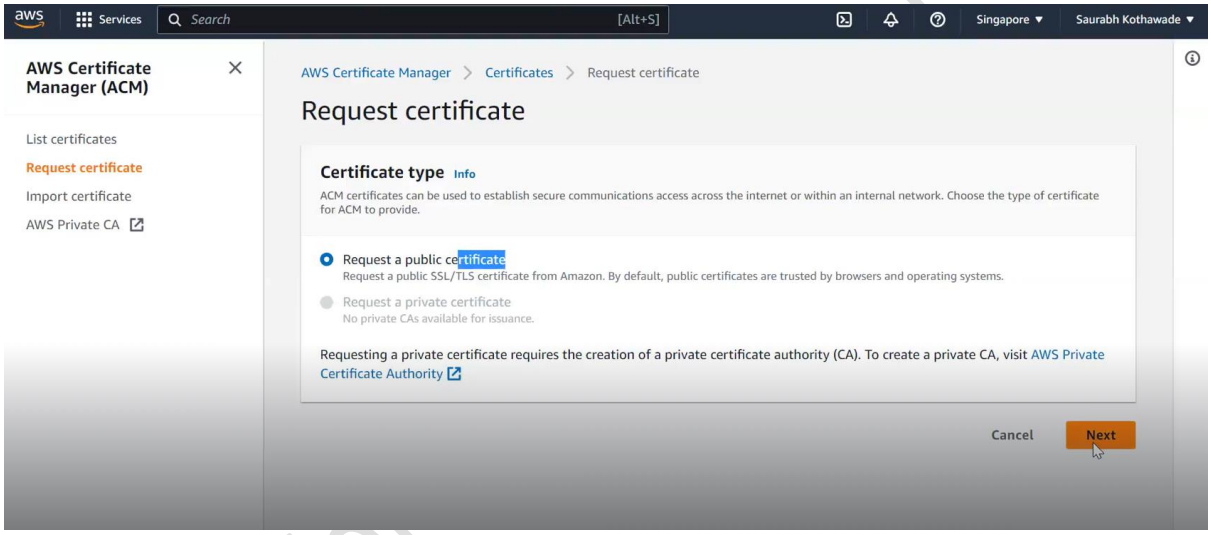


- Create an SSL certificate:

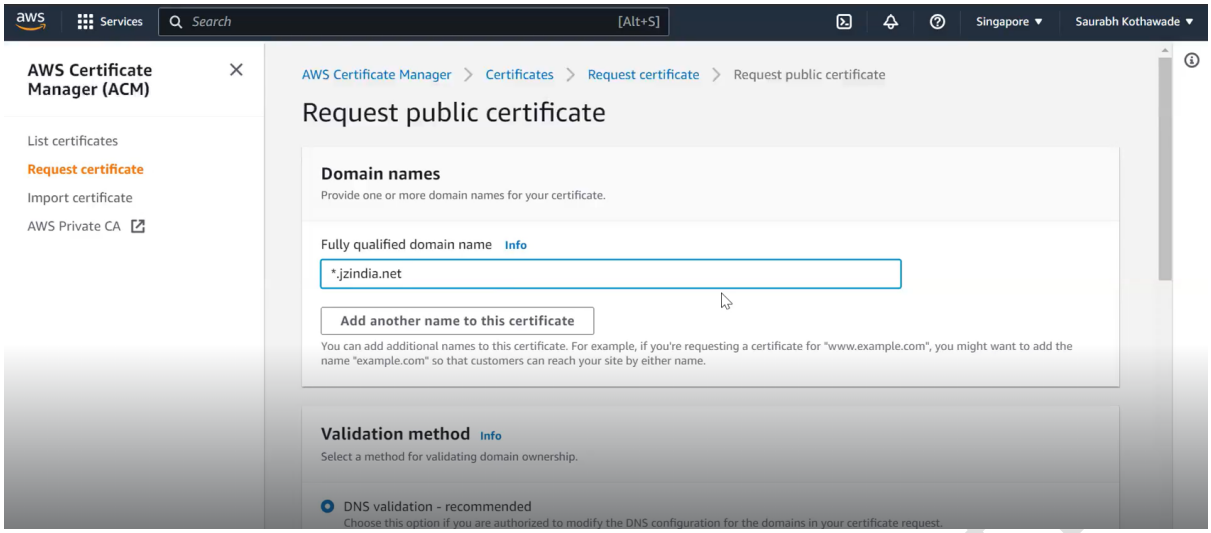


➢ Click on the "Request a certificate" button.

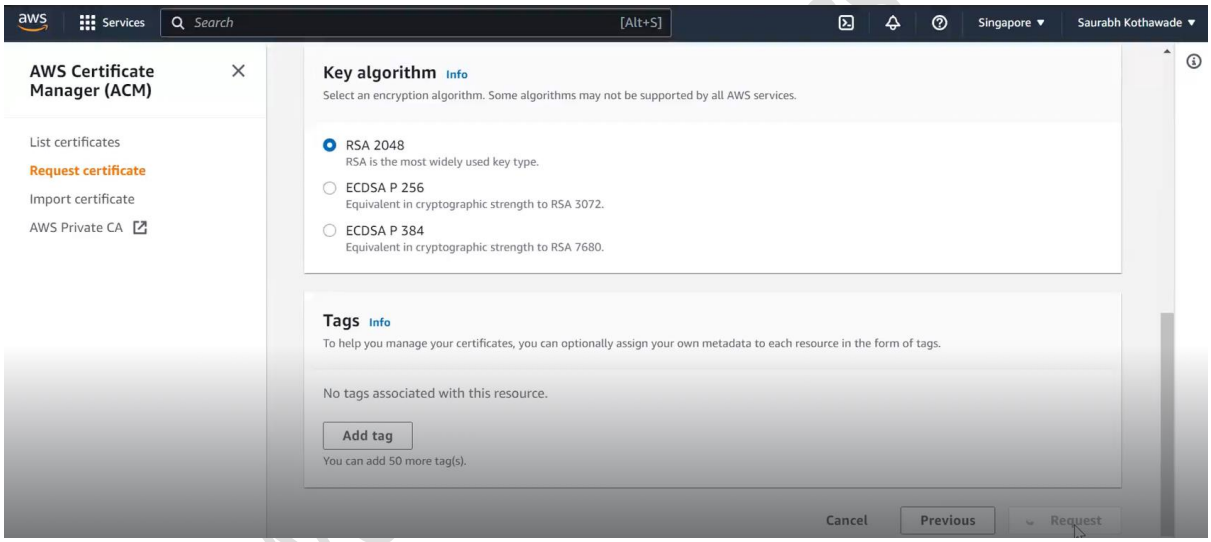




- Enter the domain name you want to secure with SSL and click on the "Next" button.



- Click on the "Request" button to submit your certificate request.



- Then after that create a record set in the “Route53” service.

- After completing the above setup you can specify that all traffic should be redirected to HTTPS.

- ***alb.ingress.kubernetes.io/listen-ports*** specifies the ports that ALB used to listen on.

- `alb.ingress.kubernetes.io/listen-ports: '[{"HTTP": 80}, {"HTTPS": 443}, {"HTTP": 8080}, {"HTTPS": 8443}]'`

- ***alb.ingress.kubernetes.io/ssl-redirect*** enables SSLRedirect and specifies the SSL port that redirects to.

- `alb.ingress.kubernetes.io/ssl-redirect: '443'`

- ***alb.ingress.kubernetes.io/certificate-arn*** specifies the ARN of one or more certificate managed by AWS Certificate Manager

- `alb.ingress.kubernetes.io/certificate-arn: arn:aws:acm:us-west-2:xxxxx:certificate/xxxxxxx`